Cloud Data Migration Playbook

Practice: Technology Advisory | Audience: Engagement Leads | Version 1.4

Overview

This playbook outlines Acme Consulting's standard approach to migrating a client's on-premises data infrastructure to a cloud platform. It is intended as a reference for engagement leads scoping a migration workstream, not as a substitute for a client-specific migration plan, which should always be produced during discovery.

Phase 1: Discovery and Assessment

Every migration engagement begins with a two-to-four week discovery phase that inventories the client's existing systems, data volumes, and integration points. A common mistake in early-stage estimates is underweighting integration complexity relative to raw data volume; a small database with twenty downstream integrations is typically a harder migration than a large database with two.

Discovery should produce a system inventory, a data classification summary noting any regulated or sensitive data categories, and a preliminary risk register. Engagements that skip a formal risk register during discovery are disproportionately likely to encounter unplanned schedule slippage during the cutover phase.

Phase 2: Migration Strategy Selection

Three strategies are commonly used, and the right choice depends on the client's tolerance for downtime and the complexity of the source system. A lift-and-shift approach moves data with minimal transformation and is fastest, but carries forward any existing schema problems. A re-platform approach modernizes the schema during migration and is the most common choice for clients with known data quality issues. A parallel-run strategy operates both systems simultaneously for a transition period and is recommended when downtime tolerance is near zero, such as for transaction-processing systems.

Phase 3: Execution

Migrations should always be executed in batches rather than as a single cutover, even for smaller datasets, since batching allows validation checkpoints between stages and limits the blast radius of any single failure. Each batch should be validated against source-system checksums before the next batch begins, and a rollback plan should exist for every batch, not only for the final cutover.

Client communication during execution matters as much as the technical work. We recommend a daily status update during active migration windows, even on days with no notable progress, since silence during a migration is one of the most common sources of client anxiety on these engagements.

Phase 4: Validation and Cutover

Before final cutover, run a full reconciliation between source and target systems covering record counts, checksums on a sampled basis, and functional testing of downstream reports or applications that depend on the migrated data. Known data quality issues identified during discovery should be remediated before cutover wherever feasible, rather than migrated as-is, since fixing them

post-migration is typically more expensive once the data is in active use.

Common Pitfalls

- Underestimating integration complexity relative to data volume during initial scoping.
- Treating discovery as optional or compressing it to save time, which increases risk later.
- Migrating known-bad data as-is instead of scheduling a cleansing sprint before cutover.
- Insufficient client communication during execution, leading to avoidable stakeholder anxiety.
- Skipping a rollback plan for intermediate batches, not just the final cutover.